1. Which tense is used to state universal scientific facts and general laws?
- **(A)** Simple Present
- **(B)** Simple Past
- **(C)** Present Continuous
- **(D)** Past Continuous

2. In 'DNA replication occurs in the 5' to 3' direction', the tense is:
- **(A)** Present Continuous
- **(B)** Simple Present
- **(C)** Present Perfect
- **(D)** Past Simple

3. Which tense is primarily used in the Materials and Methods section?
- **(A)** Simple Present
- **(B)** Simple Past
- **(C)** Present Perfect
- **(D)** Future Simple

4. In 'The cells were incubated at 37 deg C for 24 hours', the tense is:
- **(A)** Past Continuous
- **(B)** Simple Past (Passive)
- **(C)** Past Perfect
- **(D)** Present Perfect

5. Which tense connects past research discoveries to the current state of knowledge?
- **(A)** Simple Past
- **(B)** Present Perfect
- **(C)** Past Perfect
- **(D)** Present Continuous

6. Identify the present perfect tense in the following sentences:
- **(A)** We isolated the gene.
- **(B)** We have isolated the gene.
- **(C)** We had isolated the gene.
- **(D)** We were isolating the gene.

7. A Zero Conditional sentence expresses:
- **(A)** Hypothetical future outcomes
- **(B)** Universal physical/scientific truths
- **(C)** Impossible past events
- **(D)** Wishes and regrets

8. In 'If water boils, it changes into vapor', which conditional type is this?
- **(A)** Zero Conditional
- **(B)** First Conditional
- **(C)** Second Conditional
- **(D)** Third Conditional

9. A First Conditional sentence typically expresses:
- **(A)** Unreal past conditions
- **(B)** Probable future real conditions
- **(C)** Pure scientific definitions
- **(D)** Impossible fantasies

10. Choose the correct verb: 'If the enzyme is heated above 80 deg C, it ___.'
- **(A)** denatures
- **(B)** denatured
- **(C)** would denature
- **(D)** had denatured

11. Choose the correct verb: 'If you add ethanol, the nucleic acid ___.'
- **(A)** will precipitate
- **(B)** precipitated
- **(C)** would precipitate
- **(D)** would have precipitated

12. Which conditional type uses past simple in the 'if' clause and 'would + verb' in the main clause?
- **(A)** Zero Conditional
- **(B)** First Conditional
- **(C)** Second Conditional
- **(D)** Third Conditional

13. In 'If we used a different vector, the yield might improve', the type is:
- **(A)** Zero Conditional
- **(B)** First Conditional
- **(C)** Second Conditional
- **(D)** Third Conditional

14. A Third Conditional sentence expresses:
- **(A)** Past unreal/counterfactual condition
- **(B)** Present hypothetical condition
- **(C)** Real future condition
- **(D)** Universal physical fact

15. Choose the correct verb: 'If the temperature ___ lower, the crystals would not have degraded.'

(A) was
(B) were
(C) had been
(D) would be

16. In 'Figure 2 shows the protein binding kinetics', 'shows' is in:

(A) Simple Present
(B) Present Perfect
(C) Simple Past
(D) Future Simple

17. Which tense is formed by 'had + past participle'?

(A) Simple Past
(B) Past Perfect
(C) Present Perfect
(D) Past Continuous

18. In 'Before the assay ended, the control had already cleared', the tense is:

(A) Simple Past
(B) Past Perfect
(C) Present Perfect
(D) Past Continuous

19. Identify the past continuous tense:

(A) The culture grew.
(B) The culture was growing.
(C) The culture had grown.
(D) The culture has been growing.

20. Which verb form represents the present perfect continuous tense?

(A) has incubated
(B) has been incubating
(C) was incubating
(D) had been incubating

21. Stative verbs (e.g., consist, contain, belong) generally do NOT take:

(A) Simple Present
(B) Continuous (Progressive) Aspect
(C) Simple Past
(D) Perfect Aspect

22. Which of the following sentences violates stative verb rules?

(A) The plasmid contains an ampicillin resistance gene.
(B) The plasmid is containing an ampicillin gene.
(C) The plasmid contained an ampicillin gene.
(D) The plasmid has contained an ampicillin gene.

23. In formal subjunctive conditional clauses, what verb replaces 'was' for all persons?

(A) is
(B) are
(C) were
(D) be

24. Select the correct subjunctive form: 'If the inhibitor ___ present, the reaction would cease.'

(A) was
(B) were
(C) is
(D) are

25. Which auxiliary verb indicates future intention or projection in scientific research?

(A) Will
(B) Did
(C) Had
(D) Was

26. In 'By next year, the lab ___ genomic sequencing on 500 samples', the verb is:

(A) will complete
(B) will have completed
(C) completed
(D) has completed

27. What tense is 'will have completed'?

(A) Future Continuous
(B) Future Perfect
(C) Future Simple
(D) Present Perfect

28. Identify the sentence written in the simple future tense:

(A) The trial will evaluate toxicity.
(B) The trial evaluates toxicity.
(C) The trial evaluated toxicity.
(D) The trial is evaluating toxicity.

29. In 'The scientist said that the bacterium ___ Gram-negative', why is present tense retained?
- **(A)** Because it is an irregular verb
- **(B)** Because it is a timeless scientific fact
- **(C)** Because indirect speech requires present
- **(D)** Because said is present tense

30. Choose the correct conditional form: 'If the primer ___ to the template, extension occurs.'
- **(A)** anneals
- **(B)** annealed
- **(C)** would anneal
- **(D)** had annealed

31. Identify the tense: 'Researchers have been investigating CRISPR mechanisms since 2012.'
- **(A)** Present Perfect Continuous
- **(B)** Past Perfect Continuous
- **(C)** Present Continuous
- **(D)** Past Continuous

32. Which conjunction can introduce a negative conditional clause meaning 'if not'?
- **(A)** Because
- **(B)** Unless
- **(C)** Although
- **(D)** Whereas

33. Complete the sentence: 'Unless the pH is maintained at 7.4, the cells ___.'
- **(A)** will die
- **(B)** died
- **(C)** would have died
- **(D)** had died

34. In 'Had we known the sequence, we would have ordered primers', what is omitted?
- **(A)** because
- **(B)** if
- **(C)** although
- **(D)** unless

35. The construction 'Had we known...' is an example of:
- **(A)** Inverted conditional
- **(B)** Passive voice
- **(C)** Cleft sentence
- **(D)** Compound predicate

36. Choose the correct verb: 'The technician ___ the autoclave before the power failure occurred.'
- **(A)** started
- **(B)** had started
- **(C)** has started
- **(D)** is starting

37. Which tense is used in reporting specific results obtained in a completed trial?
- **(A)** Simple Present
- **(B)** Simple Past
- **(C)** Future Simple
- **(D)** Present Continuous

38. In 'Table 3 summarizes the pharmacokinetic parameters', the tense is:
- **(A)** Simple Present
- **(B)** Simple Past
- **(C)** Present Perfect
- **(D)** Future Simple

39. Choose the correct verb: 'When glucose is exhausted, yeast ___ to ethanol fermentation.'
- **(A)** switches
- **(B)** switched
- **(C)** would switch
- **(D)** had switched

40. Identify the incorrect verb form in scientific conditional sentences:
- **(A)** If the enzyme were stable...
- **(B)** If the enzyme was to be stable...
- **(C)** Were the enzyme stable...
- **(D)** Had the enzyme been stable...

41. Select the sentence with the correct tense for literature review in an Introduction:
- **(A)** Miller (1953) has synthesized amino acids.
- **(B)** Miller (1953) synthesized amino acids.
- **(C)** Miller (1953) is synthesizing amino acids.
- **(D)** Miller (1953) will synthesize amino acids.

42. Select the correct tense when referencing general scientific consensus without a specific date:
- **(A)** Many studies established that autophagy promotes survival.
- **(B)** Many studies have established that autophagy promotes survival.
- **(C)** Many studies will establish that autophagy promotes survival.
- **(D)** Many studies had established that autophagy promotes survival.

43. Identify the correct mixed conditional sentence:

(A) If we had calibrated the sensor yesterday, the data would be reliable today.
(B) If we calibrated the sensor yesterday, the data would be reliable today.
(C) If we had calibrated the sensor yesterday, the data would have been reliable today.
(D) If we calibrate the sensor yesterday, data will be reliable.

44. Choose the correct verb: 'By the time the incubation completed, the OD600 ___ 1.5.'

(A) reached
(B) has reached
(C) had reached
(D) reaches

45. Identify the erroneous tense shift in: 'We added the antibody, vortexed the tube, and then we measure absorbance.'

(A) added should be add
(B) vortexed should be vortex
(C) measure should be measured
(D) no error present

46. Choose the correct inverted conditional form for: 'If the pressure should rise beyond 2 bar...'

(A) Should the pressure rise beyond 2 bar...
(B) Rise should the pressure beyond 2 bar...
(C) Should rise the pressure beyond 2 bar...
(D) If should the pressure rise...

47. Choose the correct verb: 'If the buffer ___ prepared correctly, the protein would not precipitate now.'

(A) had been
(B) was
(C) were
(D) is

48. Select the sentence showing correct backshifting in reported speech:

(A) The investigator reported that the gene was expressed at high levels.
(B) The investigator reported that the gene expresses at high levels.
(C) The investigator reports that the gene expressed.
(D) The investigator reported that the gene will express.

49. In 'Pasteur proved that spontaneous generation ___ impossible', the verb is:

(A) was
(B) is
(C) has been
(D) had been

50. Choose the correct verb: 'We observed that cell proliferation ___ exponentially during phase two.'

(A) increased
(B) increases
(C) has increased
(D) is increasing

51. Which sentence demonstrates the future perfect tense correctly in experimental planning?

(A) The assay will be completed by Friday.
(B) The assay will have been completed by Friday.
(C) The assay is completing by Friday.
(D) The assay will complete by Friday.

52. Identify the stative verb that cannot be used in the continuous aspect:

(A) Observe
(B) Possess
(C) Measure
(D) Centrifuge

53. Select the sentence with correct conditional concord:

(A) If the PCR cycles are increased, non-specific bands appeared.
(B) If the PCR cycles are increased, non-specific bands will appear.
(C) If the PCR cycles were increased, non-specific bands will appear.
(D) If the PCR cycles had increased, non-specific bands appear.

54. Choose the correct verb: 'The team ___ on developing monoclonal antibodies for over three years.'

(A) has been working
(B) is working
(C) was working
(D) had worked

55. In 'Were the recombinant protein toxic to the host, cell growth ___ cease':

(A) will
(B) would
(C) shall
(D) can

56. Identify the sentence that correctly uses the past perfect to establish relative chronology:

(A) The cells died because the coolant leaked.
(B) The cells died because the coolant had leaked.
(C) The cells had died because the coolant leaked.
(D) The cells have died because the coolant had leaked.

57. Choose the correct verb: 'Provided that the temperature ___ constant, the enzyme follows Michaelis-Menten kinetics.'

(A) remains
(B) remained
(C) would remain
(D) had remained

58. Select the sentence with an ungrammatical conditional construction:

(A) If the cells will reach confluence, we will split them.
(B) If the cells reach confluence, we will split them.
(C) When the cells reach confluence, we will split them.
(D) Should the cells reach confluence, we will split them.

59. Why is 'If the cells will reach...' incorrect in standard English syntax?

(A) Future auxiliary 'will' is prohibited in conditional 'if'-clauses
(B) Cells is plural
(C) Confluence is uncountable
(D) Split requires past tense

60. Choose the correct verb: 'The bio-reactor ___ continuously since Monday morning.'

(A) ran
(B) is running
(C) has been running
(D) had run

61. Identify the sentence with correct sequence of tenses in a complex sentence:

(A) The researchers discovered that the virus mutated whenever heat is applied.
(B) The researchers discovered that the virus mutated whenever heat was applied.
(C) The researchers discover that the virus mutated whenever heat was applied.
(D) The researchers discovered that the virus mutates whenever heat was applied.

62. In 'Should you require additional viral vectors, please ___ the lab manager':

(A) contact
(B) contacted
(C) to contact
(D) will contact

63. Choose the correct verb: 'If the reagent ___ contaminated, the spectrophotometer readings would have been erratic.'

(A) was
(B) were
(C) had been
(D) has been

64. Select the correct verb: 'Recent investigations ___ light on the mechanism of ribosomal stalling.'

(A) shed
(B) have shed
(C) had shed
(D) are shedding

65. In 'The graph shows that as substrate concentration increases, reaction rate ___':

(A) plateaus
(B) plateaued
(C) would plateau
(D) had plateaued

66. Choose the correct conditional form: 'Had the centrifuge balanced properly, the rotor ___ not damaged.'

(A) would be
(B) would have been
(C) will be
(D) was

67. Identify the sentence that uses the present continuous tense legitimately for an ongoing experiment:

(A) The optical density is currently increasing at 0.05 units per minute.
(B) The enzyme is having high molecular weight.
(C) The plasmid is belonging to the cloning vector class.
(D) The buffer is consisting of Tris-HCl.

68. Choose the correct verb: 'Prior to our 2024 trial, no research group ___ synthesized this cyclic peptide.'

(A) has
(B) had
(C) have
(D) was

69. Select the sentence with an appropriate tense for presenting a standard mathematical equation:

(A) The Michaelis-Menten equation describes enzyme kinetics.
(B) The Michaelis-Menten equation described enzyme kinetics.
(C) The Michaelis-Menten equation was describing enzyme kinetics.
(D) The Michaelis-Menten equation has described enzyme kinetics.

70. Choose the correct verb: 'If the membrane ___ permeable, glucose would diffuse freely across the barrier.'

(A) was
(B) were
(C) is
(D) has been

71. In 'As long as the pH ___ between 6.8 and 7.2, the enzyme remains active':

(A) is maintained
(B) was maintained
(C) would be maintained
(D) had been maintained

72. Identify the incorrect use of the present perfect in historical reporting:

(A) Watson and Crick have discovered the double helix in 1953.
(B) Watson and Crick discovered the double helix in 1953.
(C) Scientists have studied the double helix since 1953.
(D) The double helix has been extensively modeled.

73. Choose the correct verb: 'By the time the conference convenes, our paper ___ in Nature Biotechnology.'

(A) will appear
(B) will have appeared
(C) appeared
(D) has appeared

74. Select the sentence where 'wish' is followed by the correct unreal past subjunctive:

(A) We wish the yield was higher.
(B) We wish the yield were higher.
(C) We wish the yield is higher.
(D) We wish the yield will be higher.

75. Choose the correct verb: 'Unless the cells ___ protected by glycerol, freezing will cause ice crystal formation.'

(A) are
(B) were
(C) will be
(D) had been

76. Identify the sentence with an inappropriate tense shift between Methodology and Discussion:

(A) We titrated the acid and find that the endpoint is neutral.
(B) We titrated the acid and found that the endpoint was neutral.
(C) Titration revealed that the endpoint is neutral.
(D) After titrating the acid, we determined the endpoint.

77. Select the sentence with a flawlessly constructed inverted counterfactual conditional:

(A) Had the secondary antibody been omitted, no false-positive fluorescence would have been observed.
(B) Had the secondary antibody omitted, no false-positive fluorescence would have been observed.
(C) If had the secondary antibody been omitted, no fluorescence would have been observed.
(D) Had been the secondary antibody omitted, no fluorescence would be observed.

78. Choose the correct verb in an unreal conditional with inverted syntax: '___ the promoter active, GFP expression would occur.'

(A) Were
(B) Was
(C) Is
(D) Had

79. Select the sentence that correctly distinguishes between general empirical fact and historical observation:

(A) Harvey discovered that blood circulates through the human body.
(B) Harvey discovered that blood circulated through the human body.
(C) Harvey has discovered that blood circulates through the human body.
(D) Harvey discovers that blood circulates through the human body.

80. Identify the sentence that violates the sequence of tenses in reported technical claims:

(A) The authors claimed that their method improves throughput by 40%.
(B) The authors claimed that their method improved throughput by 40%.
(C) The authors claim that their method improves throughput by 40%.
(D) The authors had claimed that their method had improved throughput.

81. Choose the correct verb: 'It is essential that the sterile hood ___ decontaminated before retroviral handling.'

(A) is
(B) be
(C) was
(D) are

82. The use of the base verb 'be' in 'It is essential that the hood be decontaminated' is an example of:

(A) Mandative Subjunctive
(B) Indicative Future
(C) Infinitive passive
(D) Conditional mood

83. Choose the correct verb: 'The committee recommended that Dr. Watson ___ the clinical trial director.'

(A) is
(B) become
(C) becomes
(D) became

84. Select the sentence with a grammatically correct mixed conditional representing past cause and ongoing present state:

(A) Had the viral stock not been stored at -80 deg C, its infectious titer would be negligible today.
(B) If the viral stock was not stored at -80 deg C, its titer would be negligible today.
(C) Had the viral stock not been stored at -80 deg C, its titer would have been negligible today.
(D) Were the viral stock not stored at -80 deg C, its titer is negligible today.

85. Identify the subtle tense inconsistency in this experimental abstract sentence:

(A) The extract inhibited microbial growth in vitro, but it does not protect the infected mice in vivo.
(B) The extract inhibited microbial growth in vitro, but it did not protect the infected mice in vivo.
(C) The extract inhibits microbial growth in vitro, but it did not protect the infected mice in vivo.
(D) The extract had inhibited microbial growth in vitro, but did not protect mice.

86. In 'No sooner had we initiated the reaction than the solution ___ turbid':

(A) became
(B) had become
(C) becomes
(D) was becoming

87. Choose the correct verb: 'If the enzyme were to aggregate, its catalytic efficiency ___ plummet.'

(A) will
(B) would
(C) shall
(D) can

88. Identify the sentence containing an erroneous future auxiliary inside a temporal adverbial clause:

(A) As soon as the centrifuge will stop, remove the canisters.
(B) As soon as the centrifuge stops, remove the canisters.
(C) When the centrifuge will stop, canisters must be removed.
(D) After the centrifuge will complete, take the tubes.

89. Why is 'As soon as the centrifuge will stop...' ungrammatical?

(A) Temporal clauses of time cannot take future 'will'; they take present tense
(B) Centrifuge is an inanimate noun
(C) Stop is intransitive
(D) Remove is imperative

90. Select the sentence with correct sequence of tenses regarding a superseded prior hypothesis:

(A) Earlier researchers believed that proteins were the genetic material.
(B) Earlier researchers had believed that proteins are the genetic material.
(C) Earlier researchers believe that proteins were the genetic material.
(D) Earlier researchers believed that proteins are the genetic material.

91. Choose the correct verb: 'The bioreactor ___ under aerobic conditions until the oxygen sensor malfunctioned.'

(A) operated
(B) had been operating
(C) has operated
(D) operates

92. Identify the sentence using the past subjunctive correctly in a technical hypothetical statement:

(A) If the cell membrane were permeable to sodium, osmotic lysis would inevitably follow.
(B) If the cell membrane was permeable to sodium, osmotic lysis would inevitably follow.
(C) If the cell membrane would be permeable to sodium, osmotic lysis followed.
(D) If the cell membrane is permeable to sodium, osmotic lysis would follow.

93. Select the sentence that avoids ambiguity in stating ongoing literature findings:

(A) Studies showed that CRISPR is effective.
(B) Studies have shown that CRISPR is effective.
(C) Studies had shown that CRISPR was effective.
(D) Studies show that CRISPR was effective.

94. Choose the correct verb: 'By the conclusion of this fiscal quarter, the bio-facility ___ 10,000 liters of recombinant insulin.'

(A) produces
(B) will produce
(C) will have produced
(D) has produced

95. Identify the sentence with an ungrammatical conditional inversion:

(A) Were the primers to fail, we will re-order new oligonucleotides.
(B) Were the primers to fail, we would re-order new oligonucleotides.
(C) Should the primers fail, we will re-order new oligonucleotides.
(D) Had the primers failed, we would have re-ordered new oligonucleotides.

96. Choose the correct verb: 'The principal investigator insisted that the protocol ___ followed without deviation.'

(A) is
(B) be
(C) was
(D) were

97. In 'Scarcely had the culture reached log phase when the nutrient feed ___ exhausted':

(A) became
(B) had become
(C) becomes
(D) was becoming

98. Select the sentence with correct tense usage when presenting an illustration in an academic paper:

(A) As shown in Figure 4, the mutant strain exhibited reduced pathogenicity.
(B) As shown in Figure 4, the mutant strain exhibits reduced pathogenicity.
(C) As was shown in Figure 4, the mutant strain exhibited reduced pathogenicity.
(D) As will be shown in Figure 4, the mutant strain had exhibited pathogenicity.

99. Choose the correct verb: 'If the cDNA library ___ contaminated, sequencing results would have shown chimeric reads.'

(A) were
(B) was
(C) had been
(D) has been

100. Identify the sentence that achieves flawless tense harmony, subjunctive mood, and academic scientific precision:

(A) Had the recombinant peptide folded into its native conformation, its tertiary binding cleft would have remained accessible to monoclonal targeting.
(B) Had the recombinant peptide folded into its native conformation, its tertiary binding cleft would remain accessible to monoclonal targeting.
(C) If the recombinant peptide had folded into its native conformation, its tertiary binding cleft will remain accessible.
(D) Were the recombinant peptide folded into its native conformation, its binding cleft would have remained accessible.

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Analyze the twelve-tense-aspect matrix of the English verbal system and evaluate its communicative precision in scientific discourse.

Q2. Detail the conventional tense shifts across the IMRaD sections (Introduction, Methods, Results, Discussion) of research papers.

Q3. Examine the grammatical rules and scientific applications of the Zero Conditional system.

Q4. Analyze the First Conditional system, focusing on real future conditions, probability, and modal substitutions.

Q5. Examine the Second Conditional system, highlighting hypothetical present/future states and the subjunctive 'were'.

Q6. Analyze the Third Conditional system and evaluate its utility in troubleshooting experimental failures and retrospective error analysis.

Q7. Discuss Mixed Conditionals and explain their syntactic formulation in correlating past causes with ongoing present realities.

Q8. Detail the syntax of inverted conditionals without 'if' (Had, Were, Should) and their stylistic prestige in academic publications.

Q9. Examine the Present Perfect versus Simple Past distinction in scientific citations and literature reviews.

Q10. Analyze the Past Perfect tense as an essential tool for establishing relative chronological priority in laboratory procedures.

Q11. Discuss the constraints on continuous (progressive) aspect with stative verbs in biochemistry literature.

Q12. Examine the Mandative Subjunctive in formal institutional, clinical, and laboratory safety protocols.

Q13. Analyze the sequence of tenses in reported speech and explain the 'Universal Truth' exception in biology papers.

Q14. Examine the Future-in-the-Past construction ('would' / 'was going to') in documenting historical research trajectories.

Q15. Analyze the Future Perfect and Future Continuous tenses in grant proposals and translational timeline projections.

Q16. Detail the syntax of 'Unless', 'Provided that', 'As long as', and 'In case' in biotechnology SOPs.

Q17. Examine common tense errors in the Results and Discussion sections of graduate biotechnology theses.

Q18. Analyze conditional inversion with 'Should' in laboratory emergency protocols and contingency SOPs.

Q19. Discuss the syntax and semantic difference between 'If I was' and 'If I were' in academic discourse.

Q20. Examine the Present Continuous tense: valid applications in live monitoring vs. common stylistic errors.

Q21. Analyze tense harmony in complex sentences containing adverbial time clauses ('after', 'before', 'until', 'as soon as').

Q22. Examine the syntax of 'wish' and 'if only' constructions in speculative scientific literature.

Q23. Analyze the sequence of tenses in defining equations, biological models, and software algorithms.

Q24. Detail the rules governing conditional sentences expressing concession ('even if', 'whether...or').

Q25. Analyze the past subjunctive in hypothetical comparison clauses introduced by 'as if' and 'as though'.

Q26. Examine the formation and usage of the Future Continuous tense in laboratory process automation.

Q27. Discuss subtle tense shifts when reporting historical controversies and scientific paradigm shifts.

Q28. Analyze the syntax and mechanics of conditional sentences expressing habit or typical routine behavior.

Q29. Examine errors in conditional sentence logic: the illicit insertion of 'would' in 'if'-clauses.

Q30. Analyze how choice of tense controls authorial voice, distance, and epistemic stance in scientific peer reviews.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[B]	**Q41:**	[B]	**Q61:**	[B]	**Q81:**	[B]
Q2:	[B]	**Q22:**	[B]	**Q42:**	[B]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[B]	**Q23:**	[C]	**Q43:**	[A]	**Q63:**	[C]	**Q83:**	[B]
Q4:	[B]	**Q24:**	[B]	**Q44:**	[C]	**Q64:**	[B]	**Q84:**	[A]
Q5:	[B]	**Q25:**	[A]	**Q45:**	[C]	**Q65:**	[A]	**Q85:**	[B]
Q6:	[B]	**Q26:**	[B]	**Q46:**	[A]	**Q66:**	[B]	**Q86:**	[A]
Q7:	[B]	**Q27:**	[B]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[B]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[B]	**Q88:**	[B]
Q9:	[B]	**Q29:**	[B]	**Q49:**	[B]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[B]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[B]	**Q71:**	[A]	**Q91:**	[B]
Q12:	[C]	**Q32:**	[B]	**Q52:**	[B]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[C]	**Q33:**	[A]	**Q53:**	[B]	**Q73:**	[B]	**Q93:**	[B]
Q14:	[A]	**Q34:**	[B]	**Q54:**	[A]	**Q74:**	[B]	**Q94:**	[C]
Q15:	[C]	**Q35:**	[A]	**Q55:**	[B]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[B]	**Q56:**	[B]	**Q76:**	[A]	**Q96:**	[B]
Q17:	[B]	**Q37:**	[B]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[B]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[B]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[C]
Q20:	[B]	**Q40:**	[B]	**Q60:**	[C]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

SECTION B: DETAILED SOLUTIONS FOR SUBJECTIVE QUESTIONS
1. Research articles typically follow a standard structure: Introduction, Materials and Methods, Results, Discussion, and References. Each section has a specific purpose in scientific discourse.

- Present Simple for overarching scientific claims, models, and general conclusions: "These data demonstrate that phosphorylation activates kinase activity".
- Past Simple when comparing current results with historical trials: "Our observed binding affinity was higher than the value reported by Jones (2020)".

>>Solution Question 4: Analyze the First Conditional as a tool for scientific prediction and its role in the Zero Conditional system. **Solution** The First Conditional (If + Present Simple, will + base verb) is used to express a real or likely future result that depends on a present condition. In scientific writing, it is rarely used for experimental outcomes (which are better described with the Zero Conditional) but is powerful for theoretical predictions, hypothesis testing, and discussing the implications of findings. Its function is to project forward from a known or assumed present state. **Structure and Function:** - **If + Present Simple, will + base verb:** "If the mutation disrupts the binding site, the enzyme will lose its catalytic activity." - **If + Present Simple, may/might/can + base verb:** "If the concentration exceeds the threshold, the reaction may proceed spontaneously." - **If + Present Simple, should + base verb:** "If the assay is performed correctly, the results should be reproducible." **Scientific Applications:** - **Predicting experimental outcomes:** "If we knock out the gene, the cells will not survive under these conditions." - **Discussing implications:** "If this model is accurate, it will explain the observed discrepancy between theory and experiment." - **Proposing future work:** "If we can isolate the protein, we will be able to determine its crystal structure." **Comparison with Zero Conditional:** The Zero Conditional describes general truths and established relationships ("If you heat water, it boils"). The First Conditional describes specific, future-oriented predictions ("If we heat the sample, it will boil"). In research, the Zero Conditional is more common in Results sections, while the First Conditional appears in Discussion and Conclusion sections. **Common Errors:** - Using "will" in the if-clause: "If we will increase the temperature..." (incorrect). - Using the First Conditional for general truths: "If DNA is heated, it will denature" (better as Zero Conditional). - Confusing with the Second Conditional: "If we had more time, we would repeat the experiment" (hypothetical, not a real plan). **Modal Variations:** - **If + Present Simple, may + base verb:** Expresses possibility rather than certainty. - **If + Present Simple, might + base verb:** Expresses a weaker possibility. - **If + Present Simple, can + base verb:** Expresses ability or permission. - **If + Present Simple, should + base verb:** Expresses expectation or recommendation. **Negative Forms:** - **If + Present Simple, will not (won't) + base verb:** "If the inhibitor binds, the enzyme won't function." - **If + Present Simple, may not + base verb:** "If the pH drops, the protein may not fold correctly." **Question Forms:** - **If + Present Simple, will + subject + base verb?** "If we add the reagent, will the color change?" - **If + Present Simple, should + subject + base verb?** "If the data are significant, should we repeat the analysis?" **Passive Voice:** - **If + Present Simple, will be + past participle:** "If the sample is contaminated, the results will be invalidated." **Progressive Forms:** - **If + Present Simple, will be + present participle:** "If the reaction is proceeding, we will be observing a color change." **Perfect Forms:** - **If + Present Simple, will have + past participle:** "If the experiment is successful, we will have demonstrated the mechanism." **Mixed Conditionals:** - **If + Past Simple, will + base verb:** "If we had used a different buffer, the protein will remain stable." (Past condition, present result.) **Implied Conditionals:** - **Will + base verb, if + Present Simple:** "The reaction will proceed, if the temperature is maintained." **Unless:** - **Unless + Present Simple, will + base verb:** "Unless the mutation is corrected, the protein will misfold." **As long as / Provided that:** - **As long as / Provided that + Present Simple, will + base verb:** "As long as the buffer is fresh, the enzyme will remain active." **In case:** - **In case + Present Simple, will + base verb:** "In case the sample is lost, we will repeat the extraction." **Even if:** - **Even if + Present Simple, will + base verb:** "Even if the yield is low, we will still have enough material for analysis." **Whether or not:** - **Whether or not + Present Simple, will + base verb:** "Whether or not the antibody binds, we will proceed to the next step." **Only if:** - **Only if + Present Simple, will + base verb:** "Only if the control is negative, will we trust the positive result." **Imperative:** - **If + Present Simple, imperative:** "If the solution is cloudy, discard it and prepare a new one." **Modal substitutions.**

- Moderate Probability: "If the primer anneals specifically, the amplicon should form."
- Cautious Epistemic Possibility: "If the primer anneals specifically, the amplicon may / might form."

Standard conditionals match temporal frameworks (Past-to-Past, Present-to-Future). Mixed conditionals bridge two distinct time frameworks, most commonly pairing a past counterfactual condition with an ongoing present result.

2. The Type A Mixed Conditional (Past Condition -> Present Result):
- Formula: If + [Past Perfect], ... [Would + Base Verb (Present)].
- Diagnostic Example in Biotechnology:
- Context: The lab failed to preserve an aliquot of a unique mutant strain in liquid nitrogen last year. Today, that strain is extinct.
- Mixed Formulation: "If the researchers had cryopreserved the hybridoma line in 2023, the monoclonal antibody would be available for our current clinical trials."
- Analysis: The condition is in the past ('had cryopreserved'), but the hypothetical consequence operates in the present ('would be available').

3. The Type B Mixed Conditional (Permanent Present Quality -> Past Consequence):
- Formula: If + [Past Simple / Subjunctive Were], ... [Would have + Past Participle].
- Example: "If the bacteriophage were not so lethal to Salmonella, the treatment would have failed during the outbreak."

>> Solution to Question 8: Mixed conditionals bridge two distinct time frameworks, most commonly pairing a past counterfactual condition with an ongoing present result. 2. The Type A Mixed Conditional (Past Condition -> Present Result): - Formula: If + [Past Perfect], ... [Would + Base Verb (Present)]. - Diagnostic Example in Biotechnology: - Context: The lab failed to preserve an aliquot of a unique mutant strain in liquid nitrogen last year. Today, that strain is extinct. - Mixed Formulation: "If the researchers had cryopreserved the hybridoma line in 2023, the monoclonal antibody would be available for our current clinical trials." - Analysis: The condition is in the past ('had cryopreserved'), but the hypothetical consequence operates in the present ('would be available'). 3. The Type B Mixed Conditional (Permanent Present Quality -> Past Consequence): - Formula: If + [Past Simple / Subjunctive Were], ... [Would have + Past Participle]. - Example: "If the bacteriophage were not so lethal to Salmonella, the treatment would have failed during the outbreak." ---

- Simple Past + Explicit Chronological Anchor: When a sentence specifies an author, publication year, or historical date, the Simple Past is grammatically mandatory:
- Correct: "Sanger et al. developed dideoxy chain-termination sequencing in 1977."
- Incorrect: "*Sanger et al. have developed dideoxy sequencing in 1977."
- Present Perfect + General Unanchored Consensus: When referencing ongoing discoveries or collective knowledge without anchoring them to a discrete moment:
- Correct: "Numerous laboratories have successfully adapted Sanger sequencing for high-throughput diagnostics."
- Simple Present for Enduring Theories:
- Correct: "The central dogma of molecular biology states that genetic information flows from DNA to RNA to protein."

>>Solution Question 14: Use the Past Perfect to establish a relevant historical timeline for laboratory procedures.

- Grammatical: "The buffer solution consisted of Tris-acetate."

3. The Rare Legitimate Exception (Temporary Dynamic Process):
Some stative verbs shift to dynamic meaning when indicating temporary, active changes:
- "The optical density is increasing rapidly" (Dynamic change of state).

>>Solution Question 23: Analyze the following sentence from a biochemistry paper: "The enzyme, which was discovered in 1985, is now being used in industrial processes." Explain the function of the relative clause and identify the type of relative pronoun used.

networks." (Circulates remains present simple because circulation is a permanent anatomical fact).

- Modern Example: "Mendel proved that alleles segregate independently during gamete formation." (Segregate remains present simple).

>> Solution Question Analyze the Future Perfect Past Continuous Construction (would have been doing) and translational research projects.

>>Solution Statements Questions Related to Conditional Tenses with Subject Verb Agreement and Tenses in Indirect and Direct Speech, SOPs.

- "Should the biohazard autoclave pressure exceed 2.2 bar, immediately engage the secondary emergency vent valve."
- "Should thermal runaway threaten the culture chamber, inject chilled liquid nitrogen through port B."
- "Should skin contact with acrylamide take place, flush the affected epidermis under copious running water for twenty minutes."

>>Solution Start Question Answer: The Phrase Commonly Used Is 'as soon as').

tenses in both clauses must maintain logical temporal harmony:

2. Past Reference Frame:
- Antecedent Action (Past Perfect) + Subsequent Action (Past Simple):
- "After the researcher had calibrated the pH electrode, she initiated the enzymatic titration."
- "The fermentation proceeded smoothly until the dissolved oxygen level dropped below critical threshold."
- Note: When 'before' or 'after' establishes unambiguous temporal priority, both verbs may stand in Past Simple:
- "After the researcher calibrated the pH electrode, she initiated the titration."

3. Future Reference Frame:
- Independent Clause (Future Simple) + Dependent Time Clause (Present Simple):
- Never use future tense in the time clause:
- Ungrammatical: "We will harvest the biomass as soon as the optical density will peak."
- Grammatical: "We will harvest the biomass as soon as the optical density peaks."

Solution Question 22 >>

3. Contextual Contrasts in Experimental Validation:
- When describing the historic creation of the model, use Past Simple:
- "Leonor Michaelis and Maud Menten derived this equation in 1913."
- When describing how the model behaves mathematically, use Present Simple:
- "The resulting curve asymptotically approaches Vmax."

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1. Structural Anatomy:
- Formula: Subject + will be + [Verb + -ing].
- Meaning: Indicates an action that will be in active progress at a specific designated point or across an uninterrupted window in future time.

2. Application in Industrial Bioprocess Automation:
In automated pilot plants and biopharmaceutical manufacturing schedules, the Future Continuous communicates scheduled unattended automated cycles:
- "At midnight tonight, the automated bioreactor will be performing the primary feed cycle."
- "Throughout the weekend, the robotic liquid handler will be dispensing reagents into 384-well assay plates."
- "During the third week of the clinical phase, the bio-analytical unit will be profiling pharmacokinetic metabolites."

3. Strategic Advantage:
Emphasizes seamless operational momentum and professional project management in technical Gantt-chart documentation.

>>Solution to Question 35: the tense shifts mechanism of conditional sentences and its role in verbal cognitive behavior.

- "If a new batch of competence-certified E. coli arrives, the technician immediately tests transformation efficiency with pUC19 control plasmid."
- Past Habitual:
- "During early pilot runs, whenever foaming occurred in the fermenter, technicians added silicone antifoam manually."

3. Linguistic Distinction from Zero Conditional:
While zero conditionals express immutable physical laws of nature, habitual conditionals express regular operational protocols and behavioral conventions.

>>Solution to Question 30: Here, the choice of tense centers around the distinction between 'if' clauses in scientific peer reviews.

- Past Tense + Tentative Reporting Verb (Maximum Skepticism / Distance):
- "The previous authors alleged that the peptide cured leukemia in murine models."
- Stance: Strongly implies skepticism or lack of independent replication.
